

DLMI Challenge: Fondation Models are Domain Invariant

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1. Introduction

The integration of artificial intelligence into histopathological workflows presents numerous challenges, particularly due to the inherent variability in slide preparation across different clinical centers. This heterogeneity stems from differences in staining protocols, sample preparation conditions, and scanning equipment, each of which introduces distribution shifts that can negatively impact the generalization capability of machine learning models.

To evaluate the robustness of AI systems under such conditions, DLMI Challenge proposes a binary classification task on image patches extracted from whole slide images (WSIs). The training dataset comprises samples from three medical centers, while the validation and test datasets originate from two separate, previously unseen centers. This design emphasizes the importance of domain generalization, as models must perform reliably despite changes in data distribution.

Traditional supervised learning approaches often fail to generalize effectively under such shifts, resulting in significant performance drops between training and evaluation phases. Addressing this gap necessitates strategies that are resilient to domain variability. In this study, we investigate the use of foundation models—large-scale, pretrained visual encoders—as a means to enhance generalization in histopathology. Specifically, we compare two state-of-the-art models: CTransPath (Wang et al., 2022), a hybrid convolutional-transformer model trained with contrastive learning, and UNI (Chen et al., 2024), a vision transformer pretrained with DINOv2 (Oquab et al., 2023) on a massive histology dataset.

Our experiments evaluate these models in terms of classification performance and feature invariance across domains. We perform linear probing and t-SNE-based visualization to assess the quality and domain robustness of the learned representations.

2. Related Work

Recent advances in computational pathology have focused on addressing domain shifts and limited annotations through domain adaptation techniques and the development of large-scale foundation models.

Domain Adaptation and Generalization

Histopathological images exhibit significant variability due to differences in staining protocols, scanner hardware, and tissue morphology. This variability creates domain shifts that can degrade model performance when applied across datasets or clinical sites. To address this, domain adaptation methods aim to align feature distributions between source and target domains, using techniques such as adversarial training (Zhang et al., 2019), feature normalization (Chen et al., 2021), or style transfer (Zhu et al., 2017). Generalization strategies, on the other hand, seek to learn representations that are inherently robust to such variations, often relying on large-scale, diverse training sets or regularization techniques. Despite progress, many models still suffer from performance drops when exposed to unseen domains, highlighting the need for models that generalize well across heterogeneous data sources.

Foundation Models in Histopathology

Foundation models are large-scale pretrained models designed to learn generic and transferable representations from vast amounts of unlabeled data. In computer vision, these models—often based on transformer architectures—have demonstrated strong performance across a wide range of tasks with minimal fine-tuning. Their adaptation to histopathology has been motivated by the need for data-efficient learning in a field where expert annotations are expensive and limited.

3. Methodes

CTransPath: (Wang et al., 2022) introduced **CTransPath**, a hybrid model combining a CNN encoder and a Swin Transformer for histopathological image analysis. It is pretrained in a self-supervised manner on millions of unlabeled patches from TCGA and PAIP datasets using Semantically-Relevant Contrastive Learning (SRCL), which selects semantically similar samples as positives to enhance feature diversity. CTransPath achieves competitive performance on classification and segmentation tasks and serves as a strong visual encoder for computational pathology, particularly in scenarios with limited annotations.

UNI: (Chen et al., 2024) proposed **UNI**, a self-supervised Vision Transformer (ViT-L) pretrained with DINOv2 on **Mass-100K**, a dataset of over 100 million patches from 100,000 whole-slide images across 20 tissue. UNI addresses key challenges in computational pathology, such as annotation scarcity, staining variability, and rare disease representation. It learns resolution-agnostic features without labels and achieves state-of-the-art performance on 34 downstream tasks, including slide-level classification, ROI segmentation, and few-shot learning. Compared to CTransPath and REMEDIS, UNI shows superior generalization, especially on rare cancers and high-resolution tasks, and requires fewer annotated examples. Its strong transferability and data efficiency make UNI a robust foundation model for real-world digital pathology applications.

4. Experiments

4.1. Foundation Model Setup

We evaluated two pretrained foundation models for histopathology: CTransPath (Wang et al., 2022) and UNI (Chen et al., 2024). Both models were implemented using the `timm` library and loaded from HuggingFace Hub.

4.2. Dataset and Preprocessing

The challenge data is provided in HDF5 format and includes images labeled for binary classification. All images were resized to 224×224 pixels. CTransPath used the standard ImageNet normalization and data augmentation pipeline available in `timm`, while UNI followed its official preprocessing configuration, ensuring compatibility with its pretrained weights.

4.3. Feature Extraction and Linear Probing

Each model served as a fixed feature extractor, and a lightweight linear classifier was trained on top of the extracted features. Training used binary cross-entropy loss with the Adam optimizer ($\text{lr} = 3 \times 10^{-4}$), and early stopping was applied with a patience of 10 epochs. Performance was evaluated on the validation set using classification accuracy.

4.4. Test-Time Inference

After training, the best-performing model (based on validation loss) was used to generate binary predictions on the test set. Predictions were thresholded at 0.5 and saved for submission.

4.5. Domain Robustness Evaluation with t-SNE

To qualitatively assess model generalization across acquisition centers, we conducted t-SNE visualizations of feature embeddings. For both CTransPath and UNI, features were extracted from samples spanning 4 centers (train and val), and colored by center and class.

5. Results

Model	Accuracy
DINOv2(baseline)	0.90560
CTransPath	0.95220
UNI	0.98837

Table 1: Test accuracy comparison between CTransPath and UNI.

UNI outperformed CTransPath by a notable margin on the test set. This improvement can be attributed to UNI’s large-scale pretraining on over 100 million histology patches, which likely exposed the model to greater tissue and staining variability. Additionally, UNI benefits from DINOv2 self-supervised learning, which provides resolution-agnostic and

domain-invariant feature representations. These characteristics are crucial for the DLMI challenge, where generalization across unseen domains is essential.

5.1. t-SNE Visualization of CTransPath Features

The t-SNE projection of CTransPath features (Figure 1) reveals noticeable overlaps between samples from different centers and classes. While some degree of class separation is visible, the clustering is not well-defined, especially across centers. For example, Class 0 and Class 1 samples from Center 0 and Center 1 appear entangled, indicating that CTransPath struggles to produce domain-invariant features. This behavior aligns with the observed performance gap between validation and train sets, and highlights the model’s limited generalization when exposed to strong domain shifts.

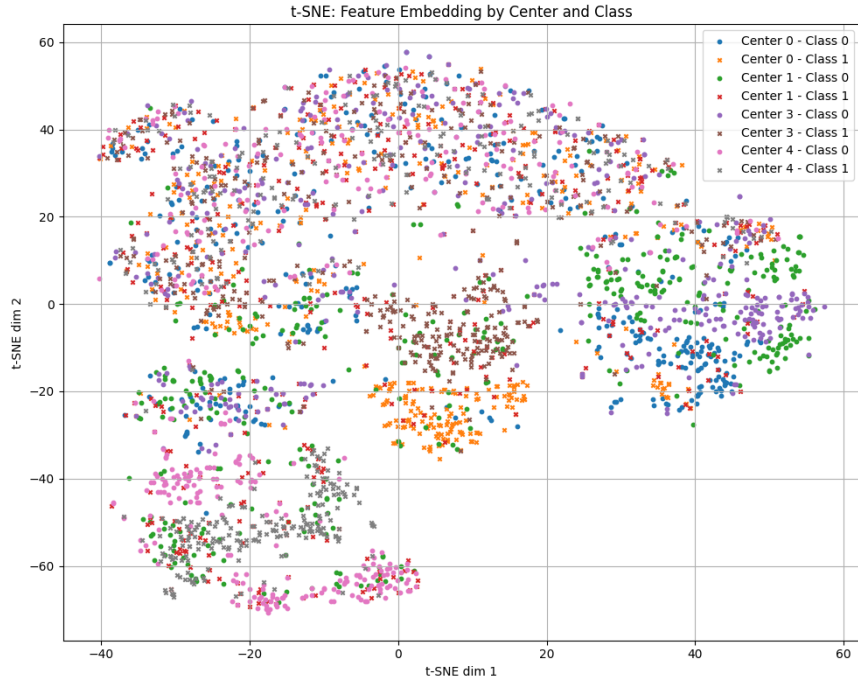


Figure 1: t-SNE plot of CTransPath feature embeddings, colored by center and class.

5.2. t-SNE Visualization of UNI Features

Figure 2 illustrates the t-SNE projection of features extracted by the UNI model. Unlike CTransPath, UNI exhibits clearer and more compact clustering by class, with samples from the same class grouping together across different centers. This structure indicates that UNI successfully learns domain-invariant features that generalize well across heterogeneous acquisition conditions. The strong separation between class 0 and class 1—despite the presence of data from five different centers—demonstrates the effectiveness of UNI’s DINOv2-based self-supervised pretraining and large-scale exposure to diverse histological patterns. These observations are consistent with UNI’s superior classification performance in the DLMI Challenge.



Figure 2: t-SNE plot of UNI feature embeddings, colored by center and class.

Comparing the two t-SNE plots highlights the advantage of UNI over CTransPath in learning more discriminative and domain-robust features. While CTransPath embeddings display overlapping regions and inconsistent class clusters across centers, UNI achieves better class separation and consistent feature groupings regardless of the data source. This indicates a higher level of generalization and confirms UNI’s suitability for handling domain shifts inherent to the DLMI challenge.

6. Conclusion

In this study, we assessed the domain generalization capabilities of two state-of-the-art foundation models—CTransPath and UNI—on the DLMI challenge. Through classification experiments and t-SNE visualizations, we demonstrated that UNI, pretrained on an extensive and diverse histopathology dataset using DINOv2, consistently outperforms CTransPath in both accuracy and feature invariance across acquisition centers. Our findings highlight the effectiveness of large-scale self-supervised pretraining for building robust, domain-invariant models in digital pathology. These results advocate for broader adoption of foundation models like UNI in clinical workflows, where variability in data acquisition remains a major obstacle to reliable AI deployment.

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